

Task 6: Create a Strong Password and Evaluate Its Strength.

- **Objective:** Understand what makes a password strong and test it against password strength tools.
- **Tools:** Online free password strength checkers (e.g., passwordmeter.com).
- **Deliverables:** Report showing password strength results and explanation.

Hints/Mini Guide:

1. Create multiple passwords with varying complexity.
2. Use uppercase, lowercase, numbers, symbols, and length variations.
3. Test each password on password strength checker.
4. Note scores and feedback from the tool.
5. Identify best practices for creating strong passwords.
6. Write down tips learned from the evaluation.
7. Research common password attacks (brute force, dictionary).
8. Summarize how password complexity affects security.

Outcome: Understanding password security and best practices.

The Password Meter

Test Your Password

Password: [REDACTED]

Hide: ☒

Score: 100%

Complexity: Very Strong

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n^4)$	16	+ 64
Uppercase Letters	Cond/Incr	$+(len-n)^2$	2	+ 28
Lowercase Letters	Cond/Incr	$+(len-n)^2$	8	+ 16
Numbers	Cond	$+(n^4)$	2	+ 8
Symbols	Flat	$+(n^6)$	4	+ 24
Middle Numbers or Symbols	Flat	$+(n^2)$	5	+ 10
Requirements	Flat	$+(n^2)$	5	+ 10

Deductions

	Type	Rate	Count	Bonus
Letters Only	Flat	$-n$	0	0

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Password Strength Checker

passwordmeter.com

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Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	16	+ 64
Uppercase Letters	Cond/Incr	$+(len-n)*2$	3	+ 26
Lowercase Letters	Cond/Incr	$+(len-n)*2$	7	+ 18
Numbers	Cond	$+(n*4)$	2	+ 8
Symbols	Flat	$+(n*6)$	4	+ 24
Middle Numbers or Symbols	Flat	$+(n*2)$	5	+ 10
Requirements	Flat	$+(n*2)$	5	+ 10
Deductions				
Letters Only	Flat	$-n$	0	0

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passwordmeter.com

Deductions	Type	Rate	Count	Bonus
Letters Only	Flat	$-n$	0	0
Numbers Only	Flat	$-n$	0	0
Repeat Characters (Case Insensitive)	Comp	-	8	- 3
Consecutive Uppercase Letters	Flat	$-(n*2)$	1	- 2
Consecutive Lowercase Letters	Flat	$-(n*2)$	5	- 10
Consecutive Numbers	Flat	$-(n*2)$	1	- 2
Sequential Letters (3+)	Flat	$-(n*3)$	0	0
Sequential Numbers (3+)	Flat	$-(n*3)$	0	0
Sequential Symbols (3+)	Flat	$-(n*3)$	0	0

Legend

- Exceptional:** Exceeds minimum standards. Additional bonuses are applied.
- Sufficient:** Meets minimum standards. Additional bonuses are applied.
- Warning:** Advisory against employing bad practices. Overall score is reduced.
- Failure:** Does not meet the minimum standards. Overall score is reduced.

Quick Footnotes

- Flat:** Rates that add/remove in non-changing increments.
- Incr:** Rates that add/remove in adjusting increments.
- Cond:** Rates that add/remove depending on additional factors.
- Comp:** Rates that are too complex to summarize. See source code for details.
- n:** Refers to the total number of occurrences.
- len:** Refers to the total password length.

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- n:** Refers to the total number of occurrences.
- len:** Refers to the total password length.
- Additional bonus scores are given for increased character variety.
- Final score is a cumulative result of all bonuses minus deductions.
- Final score is capped with a minimum of 0 and a maximum of 100.
- Score and Complexity ratings are not conditional on meeting minimum requirements.

Disclaimer

This application is designed to assess the strength of password strings. The instantaneous visual feedback provides the user a means to improve the strength of their passwords, with a hard focus on breaking the typical bad habits of faulty password formulation. Since no official weighting system exists, we created our own formulas to assess the overall strength of a given password. Please note, that this application does not utilize the typical "days-to-crack" approach for strength determination. We have found that particular system to be severely lacking and unreliable for real-world scenarios. This application is neither perfect nor foolproof, and should only be utilized as a loose guide in determining methods for improving the password creation process.

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Other sites maintained by this author: [keytonocoffee.us](#)

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